

$$\mu_1((a, b)) = |b - a|$$

a) show that $\lambda(\{a\}) = 0 \forall a \in \mathbb{R}$ $\forall F \in \mathcal{F} \setminus \{\emptyset\}$

choose $b < c$ s.t. $a \in (b, c)$ $\lambda + \mu_1 \Rightarrow \mu_1((b, c)) \geq 0$
 $= c - b$

$$\mu_1((b, c)) = \mu_1((b, a)) + \mu(\{a\}) + \mu((a, c))$$

Since μ_1
 is a measure

$$= a - b + 0 + c - a = c - b + \mu(\{a\}) > c - b$$

$$\mu_1((a, b)) = |b - a|$$

$$b) \quad \lambda((a, b)) = b - a$$

$$\lambda((a, b)) = \lambda((a, c)) + \lambda((c, b)) = b - a$$

$$\lambda([a, b)) = \lambda((a, b)) + \lambda(\{a\}) = b - a$$

$$\lambda([a, b]) = \lambda((a, b)) + \lambda(\{a\}) + \lambda(\{b\}) = b - a$$

$$\mu_1((a, b)) = |b - a|$$

$$c) \quad \text{note } (-\infty, a) \subseteq (-\infty, a], \quad (a, \infty) \subseteq [a, \infty) \\ \Rightarrow \mu((-\infty, a)) \leq \mu((-\infty, a]), \quad \mu((a, \infty)) \leq \mu([a, \infty))$$

$$(-\infty, a) = \bigcup_{n \in \mathbb{N}} [a - n, a - n + 1) \quad \text{same as above with } a \leftarrow -n$$

$$\mu((-\infty, a)) \stackrel{\mu \text{ is countable}}{\leq} \sum_{n \in \mathbb{N}} \mu([a - n, a - n + 1)) = \infty \Rightarrow \mu((-\infty, a)) = \infty$$

$$\text{analogous: } (a, \infty) = \bigcup_{n \in \mathbb{N}} (a + n - 1, a + n]$$

$$\mu((a, \infty)) = \sum_{n \in \mathbb{N}} \mu((a + n - 1, a + n]) = \sum_{n \in \mathbb{N}} 1 = \infty \Rightarrow \mu((a, \infty)) = \infty$$